Pranav Yadav

Research Interests

My research centers on computer vision for robotics, with a particular focus on agricultural robotics as a domain that demands robust perception in unstructured, real-world environments. I'm also interested in the ML/AI systems side of the field, including structured retrieval and knowledge representation methods for improving LLM reasoning, as explored in my work on HG-RAG.

Education

University of California, Merced — Merced, CA
Computer Science and Engineering — *Aug 2023 – Dec 2026*

- Key Courses: DSA, Deep Learning, NLP, Full-Stack Web Dev, Robotics, SWE, Numerical Analysis
- Dean's Honor List (Fall 2023 - Spring 2025, Spring 2026), Chancellor's Honor List (2023-26)
- IoT4Ag Summer Research Fellowship Award, Summer 2026

Experience

Robotics Intern — June 2026 – August 2026
IoT4Ag — *Atwater, CA*

- Containerized and deployed a vision-language model (Qwen3 VL 8B) in Docker on an NVIDIA Jetson AGX Orin, enabling on-device inference for real-time failure diagnosis in the agriculture Amiga robot from Farm-ng
- Built an event-driven inference pipeline triggered by ROS2 action failures, where the VLM performs visual reasoning over failure context (camera frames, error codes)
- 99% false-positive obstacle detection accuracy (200 samples) & correct action guidance accuracy of 96%

Undergraduate Research Assistant — CyberSTAR — May 2026 – June 2026
UC Merced — *Merced, CA*

- Developed a CARLA simulator training module covering environment setup, map loading, and programmatic vehicle spawning via Python scripting
- Built and documented traffic scenario generation workflows using CARLA's client-server architecture, enabling repeatable simulation environments for vehicular network research

Lab Manager — Evolab — May 2026 – Present
UC Merced — *Merced, CA*

- Oversee a lab of 10+ undergraduate researchers, coordinating weekly meetings and managing lab hours
- Lead onboarding for new members, establishing workflows and orienting them to ongoing research projects
- Collaborate directly with the PI through discussions on lab operations, research goals, and priorities

Undergraduate Research Assistant — Evolab — Feb 2025 – Present
UC Merced — *Merced, CA*

- Contributing towards building large language model support for quadruped robots by enabling a response to commands in natural language and using contextual information to make decisions (Python and ROS2)
- Experience with running participants and handling neuroimaging technology for experiments

Publications

HG-RAG: Hierarchy-Guided RAG for Structured Knowledge Graphs | *Python, NetworkX* — April 2026
https://arxiv.org/abs/2607.14095

- Designed and implemented a RAG framework that performs graph-traversal over hierarchical knowledge graphs to deliver structured context to LLMs.
- Evaluated across three world scales (18–800 nodes) and four query types against a baseline using Mistral 7B.
- HG-RAG achieved 1.86 factual accuracy vs. 0.02 baseline (2.00 scale) on large-scale graphs and demonstrated exceptional scaling on multi-hop reasoning (4.10 vs. 1.66 LLM judge score at large scale).

Projects

BeamNG Tech Autonomous Driving Research | *BeamNG, Python, Rerun* May 2026 - Present
https://github.com/Pranubot/BeamNG-Experimentation

- Built a typed Python data-capture harness over BeamNG.tech recording multi-camera RGB, LiDAR point clouds, depth maps, 6-DoF ego pose, and ground-truth 3D bounding boxes from YAML-configurable sensor rigs
- Wrote COLMAP and nuScenes-schema exporters feeding 3D Gaussian Splatting reconstruction and BEV perception pipelines, backed by unit-tested transforms across world, vehicle, OpenCV camera, and ego coordinate frames
- Shipped a CLI (doctor, record, replay, export) plus a real-time Rerun LiDAR bird's-eye viewer at 15 Hz; validated end-to-end against BeamNG.tech v0.38.3.0 on a 6-camera nuScenes-matching rig

FlowFundAI | *Next.js, React, TypeScript, Vite, Flask, Claude API, SQLite, JWT* May 2026
yeehawcuzzin.pythonanywhere.com

- SaaS finance web app for young adults and college students my group developed for our Full-Stack Web Dev Final Project.
- Parses financial documents for individual transactions and automatically pushes them to the app dashboard with Google Gemini API
- Features pluto AI assistant, powered by Claude Sonnet 4.6 refined through prompt engineering, who has the ability to access users' transaction history and spending habits for personalized guidance
- Developed with React/Vite and Flask; Our team Implemented JWT-based authentication, SQLite data persistence, and a dynamic transaction dashboard

CatTracker | *Next.js, React, TypeScript, Tailwind CSS, Google Gemini API, Google Maps API* Oct 2025
github.com/UCM-Transit-Assist/UCM-Trip-Assistant

- Built an AI-powered transit planning web app for UC Merced students using Google Gemini 2.5 Flash to process natural language queries and recommend CatTracks bus routes
- Integrated Google Maps Geocoding, Directions, and Distance Matrix APIs to calculate accurate bus paths and walking directions to bus stops
- Supported 9 bus routes with real-time schedule parsing and time-based trip planning given a user-specified arrival time

AI Resume Analysis | *Python, Node.Js, React* June 2025
ai-resume-a3733.web.app/

- Worked with UCM ML club on Web app with React and Tailwind CSS frontend and a backend done in JavaScript; Database hosted in Firebase
- Analyzes user's resume by calling to ChatGPT API with detailed prompt and guidelines for feedback
- Gives helpful feedback and calls JSearch API using keywords to find jobs near user with high relevance

Certifications

Advanced Learning Algorithms | *DeepLearning.AI, Stanford University* May 2024

- A set of 3 courses taught by Andrew Ng: Supervised Machine Learning, Regression and Classification — Unsupervised Learning, Recommenders, Reinforcement Learning — Machine Learning Specialization

Skills + Additional

Languages: Python, C++, JavaScript, HTML/CSS, SQL
ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, RAG, Pandas, NumPy, Matplotlib, YOLO, Docker
Full-Stack: React, Node.js, REST, FastAPI, Web Sockets, SQLAlchemy
Vice President and SIG AI Lead of the Association for Computing Machinery Club at UC Merced